

Pietro Carlo Boldini

Postdoctoral Researcher · Delft University of Technology

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Education

- 10/2020 – 01/2026** **PhD, Mechanical Engineering**, Delft University of Technology (NL)
Process & Energy Department, Faculty of Mechanical Engineering.
Dissertation: *Boundary-Layer Transition with Fluids at Supercritical Pressure*. Advisor: Prof. Dr. R. Pecnik. Funded by EU H2020 ERC Consolidator Grant CRITICAL (no. 864660, PI: R. Pecnik).
- 04/2016 – 12/2018** **M.Sc., Aerospace Engineering**, University of Stuttgart (DE)
GPA: 4.0/4.0 (German grade 1.2, with distinction, top 3%).
Master's Thesis (Laboratory for Advanced Simulation of Turbulence, Tsinghua University, Beijing): *Effects of Thermal Non-Equilibrium on Hypersonic Boundary-Layer Transition*. Supervisors: Prof. Dr. S. Fu, Dr.-Ing. M. J. Kloker.
- 10/2012 – 04/2016** **B.Sc., Aerospace Engineering**, University of Stuttgart (DE)
GPA: 3.7/4.0 (German grade 1.5).
Bachelor's Thesis (Institute of Space Systems): *Optimum Design of the Intake for an Atmosphere-Breathing Electric Propulsion System* — topic subsequently continued within EU H2020 project DISCOVERER (no. 737183). Supervisors: Dr.-Ing. F. Romano, Dr.-Ing. T. Binder, Prof. Dr.-Ing. G. Herdrich.

Publications

Selected Journal Articles

- [J5] P. C. Boldini, R. Hirai, B. Bugeat, and R. Pecnik. *First-Order Buoyancy Correction of Modal Instabilities in Stratified Boundary Layers*. **Journal of Fluid Mechanics**, 1036, A39, 2026 ([DOI](#)).
- [J4] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *Direct Numerical Simulation of Complete Transition to Turbulence with a Fluid at Supercritical Pressure*. **Journal of Fluid Mechanics**, 1025, A59, 2025 ([DOI](#)).
- [J3] P. C. Boldini, R. Hirai, P. Costa, J. W. R. Peeters, and R. Pecnik. *CUBENS: A GPU-Accelerated High-Order Solver for Wall-Bounded Flows with Non-Ideal Fluids*. **Computer Physics Communications**, 309, 109507, 2025 ([DOI](#)).
- [J2] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *Transient Growth in Diabatic Boundary Layers with Fluids at Supercritical Pressure*. **Physical Review Fluids**, 9, 083901, 2024 ([DOI](#)).
- [J1] B. Bugeat, P. C. Boldini, A. M. Hasan, and R. Pecnik. *Instability in Strongly Stratified Plane Couette Flow with Application to Supercritical Fluids*. **Journal of Fluid Mechanics**, 984, A31, 2024 ([DOI](#)).

Preprints

- [PP1] S. Kotturshettar and G. Alcobia and P. C. Boldini and R. Pecnik and P. Costa. *A global mass-preserving numerical method for low-Mach-number real-gas flows in closed systems*. arXiv:2607.29224, 2026. *submitted to Journal of Computational Physics* (under revision, 1st round) ([DOI](#)).

Refereed Conference Proceedings

- [C1] T. Binder, P. C. Boldini, F. Romano, G. Herdrich, S. Fasoulas. *Transmission Probabilities of Rarefied Flows in the Application of Atmosphere-Breathing Electric Propulsion*. **AIP Conference Proceedings**, 1786, 190011, 2016 ([DOI](#)).

See full list:  [Google Scholar](https://scholar.google.com/citations?user=...), [ResearchGate](https://www.researchgate.net/publication/...)

Research Experience

09/2025 – present	Postdoctoral Researcher , Aerodynamics Group, Faculty of Aerospace Engineering, Delft University of Technology (NL) Project: <i>Flow Control through Metamaterials for Transition Delay and Drag/Noise Reduction</i> — metamaterial laminar-flow control for ultra-low-drag wings, toward emission-free aviation. Supervisor: Prof. Dr. M. Kotsonis. Funded by EU Horizon Europe ERC Consolidator Grant META-WING (no. 101125132, PI: M. Kotsonis).
04/2019 – 03/2020	Research Associate , Institute of Aerodynamics and Gas Dynamics, University of Stuttgart (DE) Project: <i>Cross-flow transition control in a hypersonic boundary layer on conical geometries with thermal non-equilibrium effects</i> . Supervisor: Dr.-Ing. M. J. Kloker.
2017 – 2019	Student Research Assistant , Institute of Aerodynamics and Gas Dynamics, University of Stuttgart (DE) Project: <i>DNS of compressible turbulent boundary-layer flows</i> . Supervisor: Dr.-Ing. C. Wenzel.
2016 – 2017	Student Research Assistant , Institute of Space Systems, University of Stuttgart (DE) Project: <i>Numerical modelling for Atmosphere-Breathing Electric Propulsion</i> (EU H2020 project DISCOVERER). Supervisors: Dr.-Ing. F. Romano, Dr.-Ing. T. Binder.

Software

<i>CUBENS</i>	CUBic Equation of state Navier–Stokes solver – GPU-accelerated (NVIDIA & AMD) high-order DNS solver for wall-bounded flows with non-ideal fluids. Open source: github.com/pcbaldini/CUBENS
<i>DeHNSSo-v2</i>	Delft Harmonic Navier–Stokes Solver – nonlinear harmonic Navier–Stokes framework for stability problems in boundary layers under development during my postdoctoral work. Extended version of Westerbeek et al. (CPC, 2024, DOI): curvilinear coordinates, discrete adjoint, compressible extension, GPU acceleration. Paper in preparation.
<i>DragHeat-NIF</i>	Drag and Heat Transfer Estimation for Non-Ideal Fluids – estimates mean profiles, skin friction, and heat transfer in turbulent boundary layers with non-ideal fluids via inner/outer-layer transformations. Open source: github.com/pcbaldini/DragAndHeat-NIF

Teaching & Supervision

2023 – 2024	Teaching Assistant, <i>Advanced Heat Transfer</i> (ME45001), M.Sc. Mechanical Engineering, TU Delft.
2024	M.Sc. Thesis co-supervisor: F. Ramalho Matias, <i>Three-Dimensional Linear Stability Analysis of Flat-Plate Boundary Layers with Supercritical Fluids</i> , TU Delft.
2023	M.Sc. Thesis co-supervisor: R. Gaspar (University of Liege), <i>Investigation of the Transition to Turbulence on a Flat-Plate Boundary Layer at Supercritical Pressure</i> , TU Delft.
2023	M.Sc. Thesis co-supervisor: J. Wang (Sorbonne University), <i>Secondary instabilities in flat-plate boundary-layer flow for supercritical fluids</i> , TU Delft.
2022	M.Sc. Thesis co-supervisor: P. R. Molahalli, <i>Secondary Flows in Asymmetrical Contractions. A Numerical Study</i> , TU Delft.

Selected Talks & Presentations

[T9]	P. C. Boldini. <i>Research on boundary-layer transition with supercritical fluids</i> . DynFluid Laboratory, Arts et Metiers ParisTech, Paris, France, 2026. (Invited)
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- [T8] P. C. Boldini, R. Hirai, P. Costa, J. W. R. Peeters, and R. Pecnik. *CUBENS: A GPU-accelerated high-order solver for wall-bounded flows with non-ideal fluids*. Computer Physics Communications Seminar Series, 2025. **(Invited)**
- [T7] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *Widom-line effect on the boundary-layer transition with a highly non-ideal fluid*. 16th ERCOFTAC SIG33 Workshop, Cagliari, Italy, 2025.
- [T6] P. C. Boldini, B. Bugeat, J. W. R. Peeters, M. Kloker, and R. Pecnik. *DNS of K-type transition in a flat-plate boundary layer with supercritical fluids*. IUTAM Symposium on Laminar-Turbulent Transition, Nagano, Japan, 2024.
- [T5] P. C. Boldini, R. Hirai, P. Costa, J. W. R. Peeters, and R. Pecnik. *GPU-accelerated DNS of diabatic transitional boundary layers at supercritical pressures*. DLES 14, Erlangen, Germany, 2024.
- [T4] R. Hirai, P. C. Boldini, B. Bugeat, R. Pecnik. *DNS of H-type transition in stratified horizontal boundary layers*. 77th APS Division of Fluid Dynamics, Salt Lake City, USA, 2024. (presented by R. Hirai)
- [T3] P. C. Boldini, B. Bugeat, P. Costa, J. W. R. Peeters, and R. Pecnik. *DNS of K- and H-type transitions in a flat-plate boundary layer with supercritical fluids*. 76th APS Division of Fluid Dynamics, Washington D.C., USA, 2023.
- [T2] P. C. Boldini, B. Bugeat, P. Costa, J. W. R. Peeters, and R. Pecnik. *DNS of H-type transition in a flat-plate boundary layer with supercritical fluids*. 21st STAB-Workshop, DLR Göttingen, Germany, 2023.
- [T1] P. C. Boldini, B. Bugeat, J. W. R. Peeters, and R. Pecnik. *Transient growth in flat-plate boundary layers with a highly non-ideal fluid*. 15th ERCOFTAC SIG33 Workshop, Alghero, Italy, 2023.

Computational Skills

<i>Simulation</i>	DNS, linear/nonlinear stability analysis, OpenFOAM
<i>HPC & GPU</i>	CUDA (NVIDIA & AMD); OpenACC; MPI; large-scale HPC clusters
<i>Programming</i>	Fortran, MATLAB, Python, C++, Bash/Shell
<i>Workflow</i>	Git/GitHub; L ^A T _E X; Linux/macOS

Awards & Funding

2024	Co-author, NWO Computing Time Grant (PI: Prof. Dr. R. Pecnik), <i>Direct Numerical Simulations of Turbulent Heat Transfer with Non-Ideal Gases and Surface Roughness</i> . 30M GPU hours & 10M CPU hours on the Dutch National supercomputer Snellius (SURF NL).
2024	Co-applicant, EuroHPC JU Development Access grant (no. EHPC-DEV-2024D04-039, PI: P. Costa), <i>Towards Extreme-Scale Simulations on LUMI using CaNS</i> . 4,000 node-hours LUMI-C & 4,500 node-hours LUMI-G (CSC, Finland).
2018	DAAD scholarship for Master's Thesis research at Tsinghua University, Beijing.
2014	Top Performer Award in Astronautics I, top 5% of 111 participants, B.Sc. Aerospace Engineering, University of Stuttgart.

Professional Memberships

2023 – present	American Physical Society (APS), Member, with Division of Fluid Dynamics (DFD) affiliation.
2020 – present	J.M. Burgers Centre for Fluid Mechanics (JMBC), Registered Researcher.

Peer Review

Referee

Journal of Fluid Mechanics (4); Physical Review Fluids (1); Physics of Fluids (1); Global Power and Propulsion Society Technical Conference (1)

Delft, August 3, 2026